ME3 Voice Duplex: A Tool-Using, Full-Duplex Voice Agent for Raspberry Pi

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Abstract

ME3 is a Raspberry Pi voice-agent prototype that combines local speech processing, full-duplex audio, a provider-selectable remote language-model brain (Qwen by default, with optional GPT-5.6 Luna), and project-owned Model Context Protocol (MCP) services. The system is designed for an important interaction beyond question answering: a user can ask by voice for a web application or other coding task, and the model can call bounded file, search, command, and background-job tools to perform the work. No task-specific model training is required. Instead, the system integrates pretrained Silero VAD, faster-whisper ASR with one fixed Piper voice in the Own profile, Qwen3-TTS with one fixed CustomVoice in the Qwen profile, and a retained but inactive ElevenLabs adapter, and an OpenAI-compatible Qwen/vLLM endpoint or the OpenAI GPT-5.6 Luna API. This revision adds a plug-and-play audio profile using Qwen2-Audio-7B-Instruct ASR and Qwen3-TTS-12Hz-1.7B-CustomVoice TTS through protected vLLM listeners; the text ToolBrain and bounded tools remain shared between profiles. This paper describes the architecture, tool-call protocol, safety boundary, local verification, Raspberry Pi release workflow, and evidence boundary for the separate FULL-DUPLEX-BENCH v1.5 evaluation. The current implementation is an engineering release, not a claim of completed paper-grade overlap results. The current revision also specifies the Vocalis-Pi event contract: continuous VAD/transcript inputs, explicit turn states, interrupt controls, and an adapter seam for asynchronous MCP harness work. The workstation acceptance includes the existing local checks plus mocked request-shape and configuration tests for the Qwen audio profile. The protected DGX3 listeners were authenticated and exercised over the restored VPN route. Qwen2-Audio ASR and Qwen3-TTS both returned provider-live audio results, and a complete injected voice/tool turn passed. The latest accepted Own and Qwen releases were deployed to separate Pi directories; the live Qwen service returned HTTP 200, loaded its Qwen audio models, and reported a nonzero Fifine startup signal. A clean human spoken turn and paper-native overlap run remain separate evidence gates. The two implementation variants are now organized as complete packages under `me3 own` and `me3 qwen` are complete self-contained variants; the vendored benchmark adapter and LAN browser-speaker server were added after the previous workstation release.

Keywords

voice agents, full duplex, Raspberry Pi, tool calling, MCP, Qwen, GPT-5.6 Luna, speech interaction, edge computing

1 Introduction

Voice interfaces are often presented as a sequence: wait for the user to stop, transcribe one utterance, generate text, and play a response. A useful assistant must also manage interruption and action. A user should be able to say "build a web app" and receive evidence of files and verification, rather than a plausible paragraph describing code that was never created.

ME3 addresses these requirements with a local audio loop and a remote tool-using language model. The design separates four concerns. First, the Pi keeps voice-sensitive processing local through VAD, ASR, and TTS. Second, a Qwen model served by an OpenAI-compatible vLLM API performs dialogue reasoning by default; GPT-5.6 Luna is an optional OpenAI-compatible alternative. Third, typed tools provide an auditable action surface. Fourth, a release and deployment workflow distinguishes local preparation from actual Pi installation.

The project is intentionally not a new model-training project. The agent behavior comes from inference and orchestration: pretrained speech models, Qwen or GPT-5.6 Luna inference, MCP/tool dispatch, and a bounded local execution policy. This choice reduces iteration time and makes the engineering questions explicit.

The contributions of this work are:

- a full-duplex local voice loop with barge-in during TTS playback;
- a provider-neutral OpenAI-compatible tool loop that can create and verify a static web application through voice or text;
- one bounded registry shared by direct Qwen calls and MCP wrappers;
- confirmation-gated Twilio call/SMS and Resend email adapters with a safe provider-response boundary;
- background task supervision so builds do not block the microphone;
- a reproducible local-dev, package, deploy, and evidence workflow for the Raspberry Pi target, including a LAN browser-speaker transport and a vendored Full-Duplex-Bench v1/v1.5 adapter.

2 System Requirements and Scope

The required runtime contains four owned MCP services: VAD, ASR, TTS, and tasks. Pipecat is retained as the harness-facing integration path, while the local loop is the reference implementation used for Windows development and the Pi service. The runtime configuration is kept in one YAML file so the audio rate, speech thresholds, model choices, endpoint, tool budget, and deployment path are not duplicated across scripts.

The current engineering scope includes interruptible spoken conversation, remote Qwen inference, project-local coding tools, public web lookup, timers, system information, the existing ME2 bridge, and credential-gated outbound call, SMS, and email actions. It does not claim unrestricted Onit equivalence. An always-listening device needs a narrower default action surface than an interactive coding IDE. Capabilities that would add external side effects, credential use, repository rewrites, or arbitrary shell access must be added as reviewed typed tools.

3 Architecture

3.1 Audio and conversation path

The default audio path is:

```
microphone -> Silero VAD -> ASR -> Qwen/Luna ->
          fixed profile TTS -> speaker.
```

Capture remains active while TTS plays. The local state machine has four states: `idle`, `listening`, `thinking`, and `speaking`. Speech during `speaking` stops playback and starts a new turn. A short pre-roll is retained so the beginning of an utterance is not clipped. The VAD endpoint requires a minimum speech interval followed by configured trailing silence.

The Own profile always uses Piper `en_GB-alba-medium`; the Qwen profile always uses Qwen3-TTS CustomVoice `aiden` with a formal style instruction. Replies, backchannels, and completion announcements use the same profile voice. The Philippines-market profile keeps English, Filipino, and Taglish in the supported interaction set. The spoken policy is formal and professional: the fixed long-task acknowledgement is "I am processing your request now," and the assistant is instructed to avoid flirtatious, intimate, suggestive, teasing, or overly familiar language. The ElevenLabs adapter is retained as a documented future experiment but is not auto-selected by the active demo runtime. Qwen remains the default reasoning and tool-calling model; GPT-5.6 Luna is an optional provider using the same tool loop. ElevenLabs Realtime v1 Mini/Max is a separate hosted voice-agent architecture that also manages the LLM and ASR settings, so it is not silently substituted into this experiment.

On 26 October 2026, an authorized local provider-only test used the gitignored Windows development credential and an operator-selected account voice. ElevenLabs returned 54 raw-PCM chunks (163,468 bytes at 16 kHz), and a Filipino-language test phrase played through the workstation's default speaker. The adapter was corrected to reassemble 16-bit samples split across HTTP chunk boundaries and to classify chunked-encoding failures using the correct Requests exception handling. The account voices inspected for this test were English-labelled; therefore this result verifies the provider integration and output path, not approval of a native Filipino accent.

The duplex state machine is independent of the Qwen model. This makes the interaction control testable without a network request and leaves room for a Pipecat transport to reuse the same transition logic.

For the ME2-style Pi web experience, capture remains on the Pi while an optional browser-output transport publishes generated 16 kHz mono PCM over a LAN WebSocket. The browser schedules that stream through Web Audio and can select the PC's speaker; alternatively, the same UI selects a Pi-local output device. The HTTP dashboard exposes safe PortAudio device names and restarts only the audio loop when a route changes. A one-click browser connection is required by autoplay policy, but it does not create a push-to-talk interaction: the Pi microphone remains continuously active and barge-in is handled by the Pi state machine.

The selectable `qwen_instruct` profile changes only the ASR/TTS providers. Local Silero VAD and the same duplex state machine accept barge-in. Completed turns are sent to `Qwen/Qwen2-Audio-7B-Instruct` through an OpenAI-compatible multimodal `/v1/chat/completions` request carrying the WAV as an `audio_url` data URL. Replies are rendered by `Qwen/Qwen3-TTS-12Hz-1.7B-CustomVoice` through `/v1/audio/speech`. The text ToolBrain remains the existing `Qwen/Qwen3.8-27B` or optional Luna provider, so requests such as "build a web app" retain the same file and verification tools. Endpoint and key values are runtime-only. An ASR/TTS pair is therefore not a replacement for the general text tool-calling model.

3.2 Vocalis-Pi event contract

The implementation makes the full-duplex behavior observable through a transport-independent Vocalis-Pi orchestrator. Its external states are `LISTENING`, `THINKING`, `SPEAKING`, `BACKCHANNELING`, and `INTERRUPTED`. The local loop remains active on the microphone while the speaker plays. A VAD update received during `SPEAKING`, `THINKING`, or `BACKCHANNELING` publishes a `barge_in_event`, then emits `FLUSH_QUEUE` and `FADE_OUT` TTS controls before returning to `LISTENING`.

The protocol carries four input/event types: VAD state, transcript deltas, overlap signals, and asynchronous harness events. Its output types are assistant-text deltas, TTS control actions, turn-state transitions, and optional harness dispatch requests. A long Qwen tool turn can therefore produce a short `BACKCHANNELING` acknowledgement while its task continues. Tool progress is mirrored as sanitized harness events; arguments and results are excluded from these events because they can contain user data or credentials. The exact channel names and JSON examples are documented in the repository's Vocalis protocol manual.

The optional harness adapter emits a JSON-RPC 2.0 `tools/call` request named `harness_dispatch`. The default verified release keeps the bounded Qwen tool loop in-process for low overhead and uses the adapter as a stable seam for an Onit, Pipecat, or OpenCode background worker. This makes the contract testable without claiming that a separate external harness is already running in every deployment.

3.3 Provider-neutral tool-using brain

The remote brain is an OpenAI-compatible chat-completions client. The first request contains the recent dialogue and JSON Schema definitions for the ME3 tools. Qwen or GPT-5.6 Luna may return assistant text or a list of function calls. For every call, ME3 parses the arguments, executes the local function, appends a `role=tool` message, and asks Qwen to continue. The checked-in configuration allows 24 tool rounds for demanding multi-file or web-app tasks. After the final permitted batch, ME3 makes one no-tool summarization request instead of exposing an internal tool-limit error to the user.

For Qwen, the checked-in model is served by vLLM. For Luna, the model ID is `gpt-5.6-luna`; hidden reasoning is disabled, temperature remains omitted, and the client requests a direct response stream when supported. The protocol is illustrated below.

```
{
  "model": "Qwen/Qwen3.8-27B",
  "messages": [{"role":"user", "content":"build a web app"}],
  "tools": [{"type":"function", "function":
    {"name":"write_file", "parameters": {}}}],
  "tool_choice": "auto"
}
```

After a tool call, the result is returned as structured JSON. Qwen therefore receives the actual file path, byte count, return code, or error. The final spoken response is not accepted as evidence by itself.

3.4 Owned MCP services

The VAD server exposes frame and stream detection plus endpointing. The ASR server exposes PCM and file transcription. The TTS server exposes synthesis, streaming chunks, and installed voices. The tasks server retains timers, system information, and the ME2 bridge, and now wraps the same bounded project tool registry used by the direct Qwen path. The shared implementation avoids one policy for the local loop and another policy for MCP clients.

4 Tool Registry and Safety Boundary

Table 1 summarizes the direct functions visible to Qwen.

Table 1: Direct Qwen tools in the ME3 registry

Tool	Purpose
`list_files`	List project files and sizes.
`read_file`	Read bounded project text.
`write_file`	Create or replace a project text file.
`search_text`	Search project text files.
`run_command`	Run a direct allowlisted command.
`start_background_task`	Start a bounded build or test.
`task_status/output/cancel`	Supervise a background task.
`system_info`	Return non-secret host information.
`fetch_url/web_search`	Read public HTTP(S) resources.
`git_status`	Inspect repository status without mutation.
`channel_health`	Report safe call, SMS, and email configuration.
`send_sms/make_call/send_email`	Confirmation-gated external communication through Twilio and Resend.

The path policy resolves every path inside the ME3 project. Credential-like names, environment files, private keys, the Git directory, the virtual environment, model caches, and generated release trees are protected. Reads and writes are size bounded. Commands are tokenized and run without a shell; shell operators and patterns associated with deletion, privilege escalation, SSH/SCP, repository reset/clean/push, and package publication are rejected. Long commands run with standard input closed, captured output, and a timeout.

This boundary is a design choice rather than an implementation accident. A voice transcript can be ambiguous, and the assistant is not an appropriate place for silent arbitrary shell execution. The registry is sufficient for the demonstration web app while keeping higher-risk operations explicit.

The external communication tools use the same registry and are stricter than the local project tools. `channel_health` is network-free. Each `send_sms`, `make_call`, and `send_email` request first validates its target and returns a confirmation-required status unless the operator has explicitly confirmed the side effect. The adapters then send one bounded REST request and return only a status, channel, and safe provider ID; provider response bodies and authorization headers are discarded. Twilio call initiation additionally requires a reviewed public HTTPS TwiML URL. A complete call-center bridge, inbound webhooks, identity/CRM actions, consent, recording, and human transfer remain outside this adapter boundary.

5 Web-App Capability

The target workflow is a dependency-free web application when the user does not name a framework. Qwen writes an application directory under `agent_workspace/<name>`, lists the result, reads the important files, and runs a bounded verification command if one is appropriate. The provided demonstration contains `index.html`, `styles.css`, and `app.js`; it provides a command input, microphone placeholder, status area, and tool-activity log.

The capability is demonstrated twice: the first live request used Qwen to create the HTML file through a real tool call; a later live request used Qwen to list the directory and read

all three files. This demonstrates inference, tool dispatch, file mutation, and evidence retrieval without training a model. The demo is intentionally local and static; the current safety policy does not publish it to the public internet or install an arbitrary server.

6 Deployment Workflow

ME3 follows a four-stage local-dev workflow:

(1) **Environment gate:** import the shared Python dependencies.
(2) **Smoke gate:** load all four MCP services, call their health tools, construct the engineering benchmark, and query PortAudio when available. The vendored public fd-benchmark is also available through a direct-feed ME3 adapter; its simulation is used as a protocol smoke test, not as a hardware result.
(3) **Package:** copy the runtime files into a versioned release, write SHA-256 entries, and create a zip archive.
(4) **Deploy:** transfer the archive to the configured Pi, merge its stable `me3_voice_duplex` directory, run the idempotent installer, and record a deployment receipt.

The current target is dalmacio@cfdfnjrpi5.local at /home/dalmacio/Desktop/dandan; the Qwen profile is deployed separately at /home/dalmacio/Desktop/dandan-qwen. The separate ME2 deployment is kept outside the ME3 release. Both current receipts report `deployed` after SSH, SCP, extraction, and installation. The active Qwen service returns HTTP 200, its Qwen audio models load, and the Fifine capture path reports a nonzero startup signal. This is Pi service/audio-start evidence, not yet a clean human spoken-turn or speaker-observation claim.

The API key is never included in the archive. On the Pi it must be provisioned through a secure runtime environment. The Windows DPAPI helper can report configured/not-configured and can provide an in-memory child-process value for local testing, but its secret is not printed or written to the repository.

7 Verification and Evidence

The current local verification sequence is:

```
python scripts/dev_env_check.py
python scripts/local_smoke_test.py --with-audio
python scripts/check_paper_alignment.py
python -m pytest tests -q
python scripts/run_me3_module.py me3.agent --brain remote --text \\
  "Use system_info and summarize the host."
```

The complete test suites passed 59 tests per active variant in the current checkout. The tool-focused tests cover OpenAI-compatible schemas, blocked shell patterns, background task completion, and a mocked multi-step Qwen response. The ElevenLabs tests additionally cover split-byte PCM reassembly, genuinely odd total PCM rejection, timeout retry, and chunked-encoding retry without credential disclosure. The channel tests cover confirmation gating, E.164 validation, Twilio request construction, HTTPS call-webhook enforcement, Resend request construction, and safe metadata. Python compilation also passed. The Vocalis tests cover state transitions, JSON-RPC harness dispatch, barge-in TTS-control ordering, and LocalVoiceLoop tool-progress bridging. One current injection-only fixture run also completed with one turn and zero errors; it remains engineering evidence and not physical-speaker or v1.5 evidence. The live Qwen smoke test records an actual tool call and final answer. The local smoke gate loads the four MCP servers and confirms the current Windows PortAudio inventory. The integrated fd-benchmark simulation also completed a 10-trial development-dataset run, yielding a protocol-and-report smoke result with no ASR judge; this is not a paper-grade or physical Pi result. These are engineering checks, not a claim about Pi audio.

The evidence status at this revision is summarized in Table 2.

The deployed Pi preflight is also an explicit hardware gate. It is read-only, does not alter mixer state, and returns exit code 2 when no capture device is visible. This keeps a playback-only Pi from being mislabeled as a working full-duplex voice endpoint.

When speaker hardware is unavailable, the project uses a separate injection-only lane. The evaluator feeds recorded 16 kHz PCM into the same VAD, ASR, Qwen/tool, TTS, and interruption state machine at wall-clock frame cadence, while a virtual sink records TTS timing without opening PortAudio output. This verifies injected-input agent behavior and virtual barge-in, but does not claim physical-speaker behavior or paper-native v1.5 evidence.

To align the class evaluation with the supplied public benchmark, the release vendors the MIT-licensed fd-benchmark repository and adds `scripts/me3_fdbench_agent.py`. The adapter connects the real `LocalVoiceLoop` to the benchmark's direct-feed WebSocket: incoming 16 kHz mono int16 frames enter the same VAD/ASR/tool/TTS path, outgoing PCM is recorded by the benchmark rig, and a barge-in emits its queue-clear event. A small simulation run verifies the integration without opening a microphone or speaker; full datasets, ASR/judge scoring, and a clean Pi run remain required for publication-quality measurements.

The project also contains a separate engineering latency driver and a paper-native FULL-DUPLEX-BENCH v1.5 contract. The v1.5 contract includes four scenarios with 200, 99, 100, and 100 stimuli and response categories RESPOND, RESUME, UNCERTAIN, and UNKNOWN. The engineering five-scenario driver must not be presented as the paper's 499-utterance result. A complete captured overlap artifact, transcript classification, and prosodic before/after analysis are still required before claiming a paper-grade result.

8 Discussion

The main engineering trade-off is locality versus model capability. Keeping speech I/O local reduces the need to upload audio and makes the Pi responsive, while Qwen supplies stronger reasoning and code generation than a small local model. The cost is network dependence, endpoint latency, and the need to provision the API key on the target device.

Table 2: ME3 progress evidence at the 4 November 2026 revision

Evidence lane	Status and interpretation
Tool-capable Qwen	Verified in earlier live runs: Qwen created and inspected a web-app artifact through bounded tools.
Local duplex audio	Verified with the Windows microphone, Piper output, and two barge-in cancellations; echo can contaminate ASR on loud speakers.
LAN browser audio	The Pi-side HTTP/WebSocket server, browser PCM sink, device selection endpoint, and ME2-style SSH forwarding pass local protocol tests; physical Pi capture and LAN audio are still separate hardware gates.
Integrated fd-benchmark	Vendored MIT benchmark source and ME3 direct-feed adapter pass import/protocol checks; a 10-trial development-dataset simulation completed without ASR/judge scoring. This is not a paper-grade result.
Qwen Instruct audio profile	Adapter, runtime selection, configuration checks, and request-shape tests pass. Over the VPN, Qwen2-Audio ASR returned in 431.9 ms and Qwen3-TTS returned in 851.6 ms; the same matched run recorded faster-whisper ASR at 2732.0 ms and Piper TTS at 100.1 ms TTFB/176.7 ms total. The injected Qwen voice/tool lane completed one turn with zero errors and five shared tool identities.
ElevenLabs renderer	Verified locally: provider-live 16-kHz PCM playback through the default speaker; not a native-Filipino voice-quality result.
Qwen endpoint	Authenticated model discovery and a streamed text probe pass from the deployed Pi; a clean human spoken turn remains a separate gate.
External channels	Twilio/Resend adapters and fake-request tests pass; no live call, SMS, or email side effect has been placed.
Pi hardware	The latest Pi process loaded the Fifine capture path and reported a nonzero startup signal; a clean human spoken turn and speaker observation remain open.
Paper-native FDB v1.5	Incomplete: no clean 499-utterance overlap artifact or paper-grade result is claimed.

The typed registry makes the boundary visible and testable, but it is not a complete replacement for an interactive coding environment.

The local ElevenLabs result narrows one implementation risk: the selected streaming endpoint can produce audio in the PCM format consumed by the full-duplex playback path, and the adapter tolerates arbitrary HTTP chunk boundaries. It is not part of the active single-voice runtime. A complete voice-agent turn still requires both a reachable Qwen brain and an operator approved voice profile for the intended Filipino customer-support deployment. The new communication adapters narrow the integration boundary but do not establish that the agent can safely replace a staffed call center: provider accounts, webhooks, media streaming, consent, retention, escalation, and supervised quality gates remain necessary. Provider rendering, Qwen/tool execution, and external-channel side effects are therefore reported as separate evidence lanes rather than collapsed into one end-to-end claim.

The full-duplex behavior is also more than a transport setting. VAD must keep running during playback, the audio output must be cancellable, and the turn-state transition must preserve the interrupted user's utterance. Echo control, microphone placement, and ALSA device selection remain physical Pi concerns. A successful Windows loopback does not prove those conditions.

Future work should add a Pi-local live self-test, a reviewed HTTPS call/media-stream bridge, inbound webhook signature verification, secure credential provisioning and rotation, operator-owned Twilio/Resend tests, a reviewed web-app preview server with explicit approval, and the full v1.5 overlap capture/evaluation. Additional Onit-like capabilities should be introduced as typed, auditable functions rather than by removing the voice safety boundary.

9 Conclusion

ME3 now combines full-duplex voice interaction with a real Qwen tool loop and an explicit Vocalis-Pi event contract. A spoken build request can result in actual project files and verification, and the agent can supervise longer work without freezing the microphone. The implementation uses inference and tools rather than model training, keeps local speech services under project ownership, and records the distinction between local verification, prepared release, Pi deployment, and benchmark evidence. The ElevenLabs adapter was provider-tested locally but is not part of the active single-voice runtime. The current workstation releases pass 59 tests per variant and the injection lane passes one clean fixture run. The Pi browser-speaker path and vendored benchmark adapter pass local protocol checks, including a small simulated 10-trial run. Both accepted releases are deployed to Pi, and the Qwen Instruct audio profile is provider-live over the restored VPN route: its matched run passes both ASR and TTS, and the full injected voice/tool turn passes with zero errors. The latest prepared bundle is also Pi-installed. The next decisive checks are a clean physical Pi microphone/speaker turn, operator-owned live channel tests, and a clean captured v1.5 overlap evaluation.

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